



Assignment Cover Page

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PRN <i>VM122CS113</i>	Admission No. <i>8233</i>
Subject Name :- <i>Soft Computing</i>	Subject Code:- <i>CST444</i>
Assignment Title/No : <i>Assignment 1</i>	
Name of the faculty: <i>Asuathy Miss</i>	
Assignment Submitted on <i>...../02/26</i>	

Late submission rules : Max mark will reduced to 50% for 1-5 working day's delay, no mark will be awarded thereafter.

I am hereby confirming that this assignment is my own and I haven't adopted any unfair means in any steps of its preparation to enhance my performance in this assignment. *hija Bose*

Date : *...../02/26*

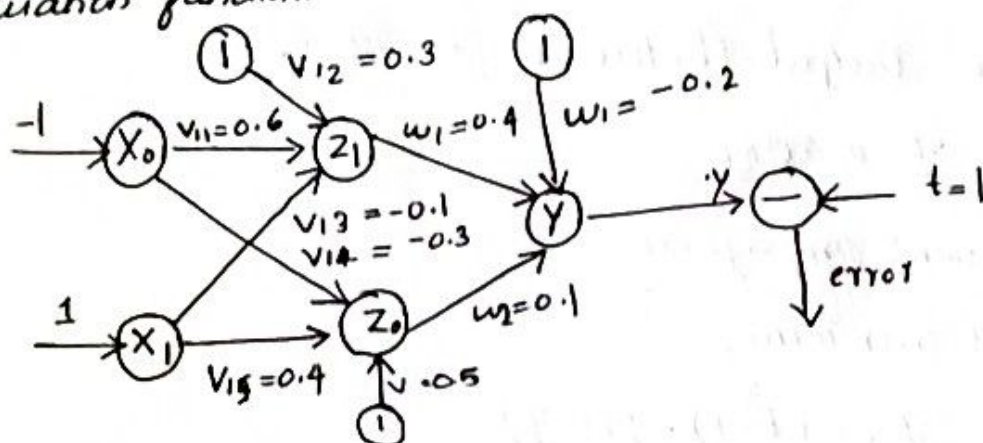
hija Bose
Sign with Name

Assignment subdivision	Maximum Mark	Marks awarded	Remarks
A			
B			
C			

Feed back/suggestions :

Name and sign of the faculty

- 7) Using the back-propagation network, find the new weights for the network shown in the figure. It is presented with the input pattern $[-1, 1]$, and the target Output is 1. Use a learning rate $\alpha = 0.05$ and binary Sigmoidal activation function.



1) Initial Weights

$$\begin{aligned} \text{weight : bias} \rightarrow Z_1 &= 0.3 \\ \text{bias} \rightarrow Z_2 &= 0.5 \end{aligned}$$

2) forward propagation
Hidden layer

For Z_1

$$\text{net } z_1 = (-1)(0.6) + (1)(-0.1) + (1)(0.3)$$

$$\text{net } z_1 = -0.6 - 0.1 + 0.3 = \underline{\underline{-0.4}}$$

$$Z_1 = \sigma(-0.4) = \underline{\underline{0.4013}}$$

For Z_2

$$\text{net } z_2 = (-1)(-0.4) + (1)(0.4) + (1)(0.5)$$

$$\text{net } z_2 = 0.4 + 0.4 + 0.5 = 1.3$$

$$Z_2 = \sigma(1.3) = \underline{\underline{0.7858}}$$

Output layer.

$$\text{net } y = (0.4013)(0.4) + (0.7858)(0.1) + (1)(0)$$

$$\text{net } y = 0.1605 + 0.0786 + 0.2$$

$$\text{net } y = \underline{\underline{0.4391}}$$

Correct Output (Before weight Output)

$$y = 0.6081$$

3) Backward propagation

Output Error Term

$$\delta y = (t - y) \cdot y(1 - y)$$

$$\delta y = (1 - 0.6081)(0.6081)(0.3919)$$

$$\delta y = 0.0934$$

4) Update Hidden $\rightarrow$ Output Weights

$$\Delta w = \alpha \delta y z$$

$$w_{21y}$$

$$\Delta w = 0.25(0.0934)(0.4013)$$

$$= 0.00937$$

$$w_{21y}^{\text{new}} = \underline{\underline{0.4094}}$$

$$w_{22y}$$

$$\Delta w = 0.25(0.0934)(0.7858) = 0.01836$$

$$w_{22y}^{\text{new}} = \underline{\underline{0.1184}}$$

Bias to y

$$\Delta b = 0.25 (0.0934) = 0.02335$$

$$b_y^{\text{new}} = \underline{\underline{0.2234}}$$

5) Hidden Layer Error Terms

$$\delta_2 = z(1-z)\delta_y w$$

for z_1

$$\delta_{z1} = (0.4013)(0.5987)(0.0934)(0.4)$$

$$\delta_{z1} = \underline{\underline{0.00898}}$$

for z_2

$$\delta_{z2} = (0.7858)(0.2142)(0.0934)(0.1)$$

$$\delta_{z2} = \underline{\underline{0.001572}}$$

6) Update Input $\rightarrow$ Hidden weights

$$\Delta V = a \delta_2 x$$

Updated weights for z_1

weight

New Value

V_{x1z1}

0.5978

V_{x2z1}

-0.0978

Bias z_1

0.3023

updated weights for z_1

<u>weight</u>	<u>New Value</u>
V_{x1z2}	-0.40039
V_{x2z2}	0.40039
Bias z_2	0.50039

1) Network Output

$$y = 0.6081$$

2) Updated weights

Hidden $\rightarrow$ Output

$$\bullet w_{z1y} = 0.4094$$

$$\bullet w_{z2y} = 0.1184$$

$$\bullet b_y = 0.2234$$

Input $\rightarrow$ Hidden

$$\bullet V_{x1z1} = 0.5918$$

$$\bullet V_{x2z1} = -0.0978$$

$$\bullet b_{z1} = 0.3023$$

$$\bullet V_{x1z2} = -0.40039$$

$$\bullet V_{x2z2} = 0.40039$$

$$\bullet b_{z2} = 0.50039$$